PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GORVERNMENT

LONGIDO DISTRICT COUNCIL

NAMANGA SECONDARY SCHOOL

FORM THREE WEEKLY TEST 3. AUGUST 2023.

CHEMISTRY 1

INSTRUCTIONS TIME: 1:30 HRS

i. This paper consists of three (**3**) questions. Answer **ALL** questions properly

ii. Arrange your work clearly.

1. a. Read the table below and answer the questions that follows

EXPERIMENT	TIME TAKEN, t(S)	PHYSICAL STATES OF THE SPECIES	CONDITION OF THE WALL OF THE BEAKER CONTAINING MIXTURES	TOTAL NUMBER OF MOLES OF REACTANT(S)	TOTAL NUMBER OF MOLES OF PRODUCT(S)
A	10	Gases	Hot	4	2
B	5	Solid and gases	Cold	1	4
C	12	gases	Hot	2	1
D	14	Solid, aqueous and gases	cold	3	2

Questions **(14marks)**

i. arrange the experiments above in order of increasing the rate of chemical reaction

ii. identify experiments which are exothermic and which are endothermic.

iii. Identify the experiment whose rate of reaction can be affected by pressure.

b. If the experiments above were reversible reactions, identify the experiment whose position of equilibrium lies more forward if

i. Temperature is increased

ii. Temperature is lowered

iii. Pressure is increased **(8marks)**

iv. Pressure is decreased

2. a. Define the following terms

i. electrolysis

ii. electrode

iii. anode (**8marks**)

iv. cathode

b. i. Mention three factors affecting the selective discharge of ions on electrode during electrolysis (**12marks**)

ii. Mention three factors affecting the magnitude of a substance liberated at electrode during electrolysis**.** **(9marks)**

iii. State Faraday's first law of electrolysis. (**3marks**)

c. Copper (II) sulphate solution undergo electrolysis by using carbon electrodes

i. write the chemical formula of the electrolyte used (**1marks**)

ii. mention four important materials present in the solution (**4marks**)

iii. show the reactions taking place at each electrode and identify the main product on cathode and anode. (**8marks**)

iv. write the overall ionic equation for the electrolysis above (**5marks**)

3. a. Define the following terms as used in extraction of metals

i. metals
ii. minerals
iii. ore
iv. physical strength of metals
v. chemical strength of metals
vi. reducing agent (**20marks**)
vii. oxidizing agent
viii. reduction
ix. oxidation
x. extraction of metals

b. Kelvin won a price for successful sorting of metals from a box containing various materials. Mention any four physical characteristics used by Kelvin until he succeeds to sort the materials. (**8marks**)

"***WORKING HARD PLUS CONFIDENCE IN YOUR WORKINGS MINUS PUNIC EQUALS TO SUCCESS''***